STUDYLAB HUB · COLLEGE PHYSICS I & AP PHYSICS · UNIT 1 (CHAPTERS 1–4)

Unit 1: 1D & 2D Kinematics & Projectile Motion

Displacement, Velocity & Acceleration · Motion Graphs · Kinematic Equations · Vector Components & Projectiles

College Physics I & AP Physics — Unit 1 (Chapters 1–4): Kinematics in 1D & 2D (Projectile Motion & Vectors)

Independent, original study guide for College Physics I & AP Physics (informed by OpenStax open textbooks, CC BY 4.0).

1. Core Unit Roadmap

Section	Core Principle & Definitions	High-Yield Exam Application
Part 1 · Fundamental Concepts	Displacement, Velocity & Acceleration	Identifying core variables, boundary conditions, and conserved quantities
Part 2 · Quantitative Models & Mechanisms	Motion Graphs	Applying governing equations and predicting how system state changes under perturbation
Part 3 · Advanced Problem Solving	Vector Components & Projectiles	Avoiding common sign, unit, and conceptual traps on multiple-choice and free-response questions

2. Essential Principles & Step-by-Step Analysis

1. **First-Principles Mechanism**:

- Every problem in **Unit 1 (Chapters 1–4): Kinematics in 1D & 2D (Projectile Motion & Vectors)** reduces to identifying the underlying structure, forces, rates, or invariants before plugging in numbers.
- Always state the governing principle explicitly at the start of any free-response justification.

2. **High-Yield Comparison & Edge Cases**:

- Distinguish between equilibrium/steady-state behavior and transient/limiting behavior.
- Verify dimensional consistency and limiting cases (e.g., as distance or concentration approaches zero or infinity).

3. **Worked Exam Strategy**:

- Write out the known quantities with units, select the exact relation connecting them, solve symbolically first, and then substitute numerical values.